PyQt5 Widgets

Using QPushButton, QCheckBox, QComboBox, QLabel and QSlider widgets

by Martin Fitzpatrick Last updated Jan 21 PyQt5 Getting started with PyQt5

PyQt5 Tutorial — Getting started with PyQt5

This tutorial is also available for PySide6 , PyQt6 and PySide2

In Qt, like in most GUI frameworks, **widget** is the name given to a component of the UI that the user can interact with. User interfaces are made up of multiple widgets, arranged within the window.

Qt comes with a large selection of widgets available and even allows you to create your own custom and customized widgets. In this tutorial, you'll learn the basics of some of the most commonly used widgets in Qt GUI applications.

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A Quick Widgets Demo

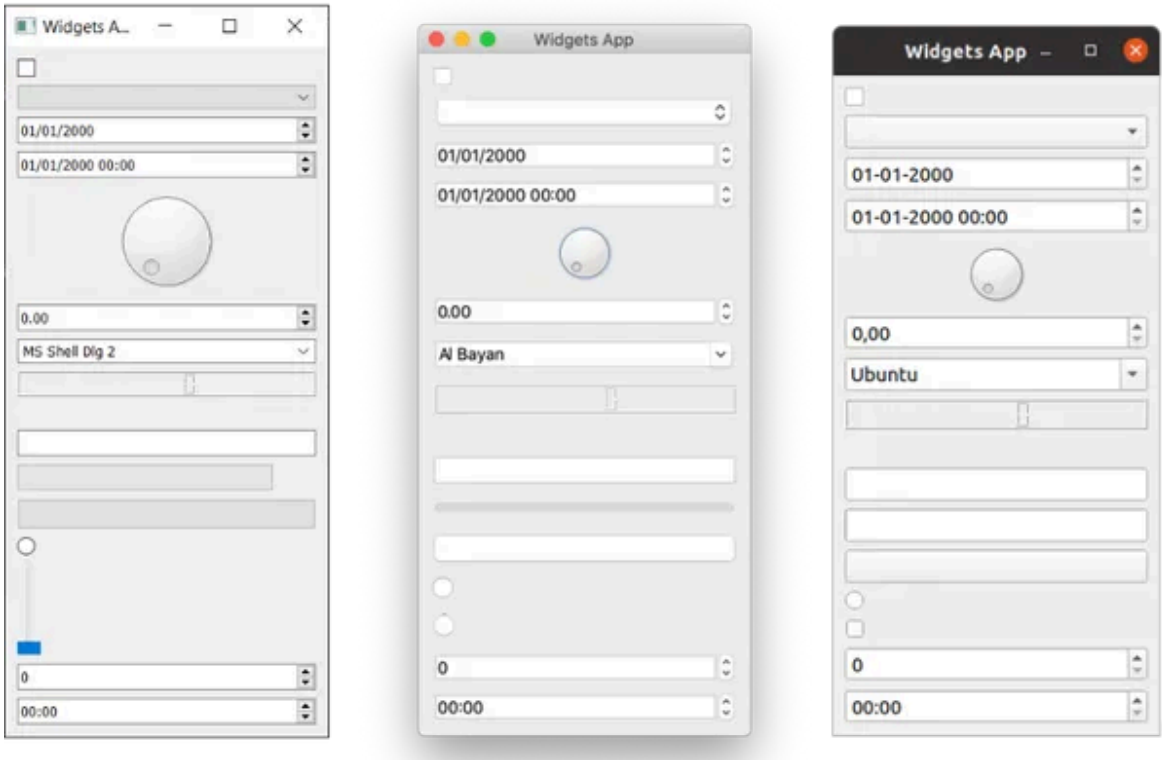
First, let's have a look at some of the most common PyQt widgets. The following code creates a range of PyQt widgets and adds them to a window layout so you can see them together:

PYTHON

```
import sys

from PyQt5.QtWidgets import (
    QApplication,
    QCheckBox,
    QComboBox,
    QDateEdit,
    QDateTimeEdit,
    QDial,
    QDoubleSpinBox,
    QFontComboBox,
    QLabel,
    QLCDNumber,
    QLineEdit,
    QMainWindow,
    QProgressBar,
    QPushButton,
    QRadioButton,
    QSlider,
    QSpinBox,
    QTimeEdit,
```

```
    QVBoxLayout,
    QWidget,
)

# Subclass QMainWindow to customize your application's main window
class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()

        self.setWindowTitle("Widgets App")

        layout = QVBoxLayout()
        widgets = [
            QCheckBox,
            QComboBox,
            QDateEdit,
            QDateTimeEdit,
            QDial,
            QDoubleSpinBox,
            QFontComboBox,
            QLCDNumber,
            QLabel,
            QLineEdit,
            QProgressBar,
            QPushButton,
            QRadioButton,
            QSlider,
            QSpinBox,
            QTimeEdit,
        ]

        for w in widgets:
            layout.addWidget(w())

        widget = QWidget()
        widget.setLayout(layout)

        # Set the central widget of the Window. Widget will expand
        # to take up all the space in the window by default.
        self.setCentralWidget(widget)

app = QApplication(sys.argv)
window = MainWindow()
window.show()
app.exec()
```

Run it! You'll see a window appear containing all the widgets we've created:



Big ol' list of widgets on Windows, Mac & Ubuntu Linux.

We'll cover how layouts work in Qt in the next tutorial.

Let's have a look at all the example widgets, from top to bottom:

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Widget	What it does
QCheckbox	A checkbox
QComboBox	A dropdown list box
QDateEdit	For editing dates and datetimes
QDateTimeEdit	For editing dates and datetimes
QDial	Rotatable dial
QDoubleSpinbox	A number spinner for floats

Widget	What it does
QFontComboBox	A list of fonts
QLCDNumber	A quite ugly LCD display
QLabel	Just a label, not interactive
QLineEdit	Enter a line of text
QProgressBar	A progress bar
QPushButton	A button
QRadioButton	A toggle set, with only one active item
QSlider	A slider
QSpinBox	An integer spinner
QTimeEdit	For editing times

There are far more widgets than this, but they don't fit so well! You can see them all by checking the Qt documentation.

Next, we'll step through some of the most commonly used widgets and look at them in more detail. To experiment with the widgets, we'll need a simple application to put them in. Save the following code to a file named `app.py` and run it to make sure it's working:

PYTHON

```
import sys

from PyQt5.QtCore import Qt
from PyQt5.QtGui import QPixmap
from PyQt5.QtWidgets import (
    QApplication,
    QCheckBox,
    QComboBox,
    QDoubleSpinBox,
    QLabel,
    QLineEdit,
    QListWidget,
    QMainWindow,
    QSlider,
    QSpinBox,
)

class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()

        self.setWindowTitle("My App")

app = QApplication(sys.argv)
```

```
window = MainWindow()
window.show()
app.exec()
```

In the code above, we've imported a number of Qt widgets. Now we'll step through each of those widgets in turn, adding them to our application and seeing how they behave.

QLabel

We'll start the tour with `QLabel`, arguably one of the simplest widgets available in the Qt toolbox. This is a simple one-line piece of text that you can position in your application. You can set the text by passing in a string as you create it:

PYTHON

```
widget = QLabel("Hello")
```

You can also set the text of a label dynamically, by using the `setText()` method:

PYTHON

```
widget = QLabel("1")  # The label is created with the text 1.
widget.setText("2")   # The label now shows 2.
```

You can also adjust font parameters, such as the size of the font or the alignment of text in the widget:

PYTHON

```
class MainWindow(QMainWindow):

    def __init__(self):
        super().__init__()

        self.setWindowTitle("My App")

        widget = QLabel("Hello")
        font = widget.font()
        font.setPointSize(30)
        widget.setFont(font)
        widget.setAlignment(Qt.AlignHCenter | Qt.AlignVCenter)

        self.setCentralWidget(widget)
```

QLabel on Windows, Mac & Ubuntu Linux.

Font tip Note that if you want to change the properties of a widget font it is usually better to get the *current* font, update it, and then apply it back. This ensures the font face remains in keeping with the desktop conventions.

The alignment is specified by using a flag from the `Qt` namespace. The flags available for horizontal alignment are listed in the following table:

Flag	Behavior
Qt.AlignLeft	Aligns with the left edge.
Qt.AlignRight	Aligns with the right edge.
Qt.AlignHCenter	Centers horizontally in the available space.
Qt.AlignJustify	Justifies the text in the available space.

Similarly, the flags available for vertical alignment are:

Flag	Behavior
Qt.AlignTop	Aligns with the top.
Qt.AlignBottom	Aligns with the bottom.
Qt.AlignVCenter	Centers vertically in the available space.

You can combine flags together using pipes (`|`). However, note that you can only use vertical or horizontal alignment flags at a time:

```python
align_top_left = Qt.AlignLeft | Qt.AlignTop
```

Note that you use an *OR* pipe (`|`) to combine the two flags (not `A &amp; B`). This is because the flags are non-overlapping bitmasks. For example, `Qt.AlignmentFlag.AlignLeft` has the hexadecimal value `0x0001` , while `Qt.AlignmentFlag.AlignBottom` is `0x0040` . By ORing them together, we get the value `0x0041` , representing 'bottom left'. This principle applies to all other combinatorial Qt flags. If this is gibberish to you, then feel free to ignore it and move on. Just remember to use the pipe (`|`) symbol.

Finally, there is also a shorthand flag that centers in both directions simultaneously:

Flag	Behavior
`Qt.AlignCenter`	`Centers horizontally *and* vertically.`

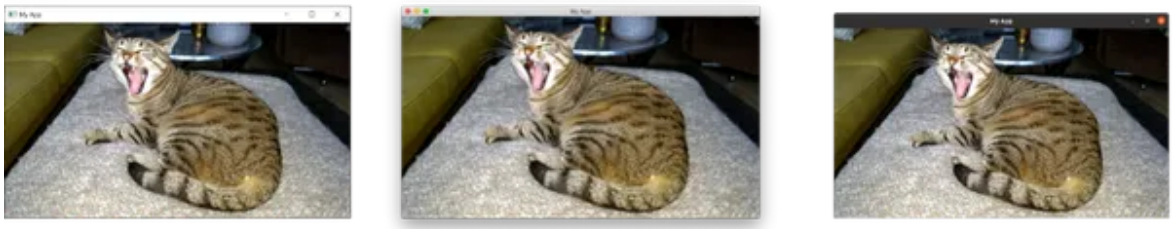
Weirdly, you can also use `QLabel` to display an image using `setPixmap()` . This accepts a *pixmap*, which you can create by passing an image filename to the `QPixmap` class.

Below is an image which you can download for this example.

"Otje" the cat.

Place the file in the same folder as your code, and then display it in your window as follows:

```python
widget.setPixmap(QPixmap('otje.jpg'))
```



"Otje" the cat, displayed in a window.

What a lovely face. By default, the image scales while maintaining its aspect ratio. If you want it to stretch and scale to fit the window completely, then you can call `setScaledContents(True)` on the `QLabel` object:

```python
widget.setScaledContents(True)
```

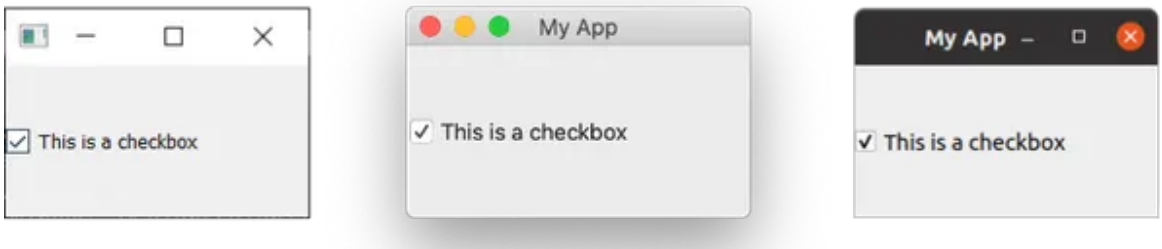
This way, your image will stretch and scale to fit the window completely.

QCheckBox

The next widget to look at is `QCheckBox()`, which, as the name suggests, presents a checkable box to the user. However, as with all Qt widgets, there are a number of configurable options to change the widget's default behaviors:

```python
class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()

        self.setWindowTitle("My App")

        widget = QCheckBox("This is a checkbox")
        widget.setCheckState(Qt.Checked)

        # For tristate: widget.setCheckState(Qt.PartiallyChecked)
        # Or: widget.setTriState(True)
```

```
        widget.stateChanged.connect(self.show_state)

        self.setCentralWidget(widget)

    def show_state(self, s):
        print(s == Qt.Checked)
        print(s)
```



QCheckBox on Windows, Mac & Ubuntu Linux.

You can set a checkbox state programmatically using the `setChecked()` or `setCheckState()` methods. The former accepts either `True` or `False`, which correspond to the checked or unchecked states, respectively. However, with `setCheckState()`, you also specify a particular checked state using a `Qt` namespace flag:

Flag	Behavior
`Qt.Unchecked`	`Item is unchecked`
`Qt.PartiallyChecked`	`Item is partially checked`
`Qt.Checked`	`Item is checked`

A checkbox that supports a partially-checked (`Qt.PartiallyChecked`) state is commonly referred to as 'tri-state', which is being neither on nor off. A checkbox in this state is commonly shown as a greyed-out checkbox, and is commonly used in hierarchical checkbox arrangements where sub-items are linked to parent checkboxes.

If you set the value to `Qt.PartiallyChecked` the checkbox will become tristate. You can also set a checkbox to be tri-state without setting the current state to partially checked by using `setTriState(True)`

You may notice that when the script is running, the current state number is displayed as an `int` with checked = `2`, unchecked = `0`, and partially checked = `1`. You don't need to remember these values, the `Qt.Checked` namespace variable `== 2`, for example. This is the value of these state's respective flags. This means you can test state using `state == Qt.Checked`.

QComboBox

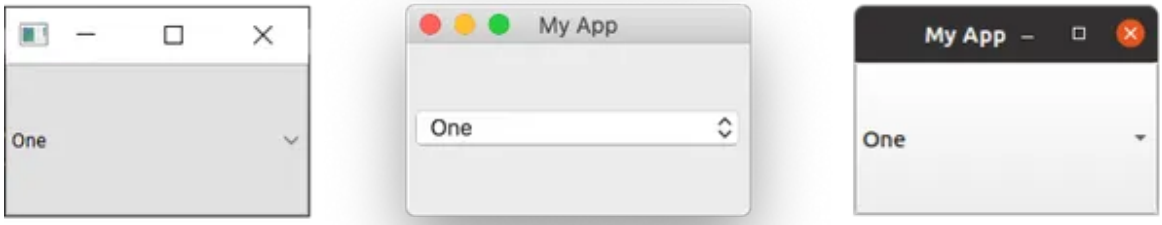
The `QComboBox` is a drop-down list, closed by default with an arrow to open it. You can select a single item from the list, with the currently selected item being shown as a label on the widget. The combo box is suited for the selection of a choice from a long list of options.

> You have probably seen the combo box used for the selection of font face, or size, in word processing applications. Although Qt actually provides a specific font-selection combo box as `QFontComboBox`.

You can add items to a `QComboBox` by passing a list of strings to `addItems()`. Items will be added in the order they are provided:

PYTHON

```
class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()

        self.setWindowTitle("My App")

        widget = QComboBox()
        widget.addItems(["One", "Two", "Three"])

        # Sends the current index (position) of the selected item.
        widget.currentIndexChanged.connect( self.index_changed )

        # There is an alternate signal to send the text.
        widget.currentTextChanged.connect( self.text_changed )

        self.setCentralWidget(widget)

    def index_changed(self, i): # i is an int
        print(i)

    def text_changed(self, s): # s is a str
        print(s)
```



QComboBox on Windows, Mac & Ubuntu Linux.

The `currentIndexChanged` signal is triggered when the currently selected item is updated, by default passing the index of the selected item in the list. There is also a `currentTextChanged` signal, which instead provides the label of the currently selected item, which is often more useful.

`QComboBox` can also be editable, allowing users to enter values not currently in the list and either have them inserted or simply used as a value. To make the box editable, use the `setEditable()` method:

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```python
widget.setEditable(True)
```

You can also set a flag to determine how the insertion is handled. These flags are stored on the `QComboBox` class itself and are listed below:

Flag	Behavior
QComboBox.NoInsert	Performs no insert.
QComboBox.InsertAtTop	Inserts as first item.
QComboBox.InsertAtCurrent	Replaces the currently selected item.

Flag	Behavior
QComboBox.InsertAtBottom	Inserts after the last item.
QComboBox.InsertAfterCurrent	Inserts after the current item.
QComboBox.InsertBeforeCurrent	Inserts before the current item.
QComboBox.InsertAlphabetically	Inserts in alphabetical order.

To use these, apply the flag as follows:

```python
widget.setInsertPolicy(QComboBox.InsertAlphabetically)
```

You can also limit the number of items allowed in the box by using the `setMaxCount()` method:

```python
widget.setMaxCount(10)
```


For a more in-depth look at the `QComboBox`, check out our QComboBox documentation.

QListWidget

This widget is similar to `QComboBox`, except options are presented as a scrollable list of items. It also supports the selection of multiple items at once. A `QListWidget` offers a `currentItemChanged` signal, which sends the `QListWidgetItem` (the element of the list widget), and a `currentTextChanged` signal, which sends the text of the current item:

```python
class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()

        self.setWindowTitle("My App")

        widget = QListWidget()
        widget.addItems(["One", "Two", "Three"])

        widget.currentItemChanged.connect(self.index_changed)
```

```
        widget.currentTextChanged.connect(self.text_changed)

        self.setCentralWidget(widget)

    def index_changed(self, i): # Not an index, i is a QListWidgetItem
        print(i.text())

    def text_changed(self, s): # s is a str
        print(s)
```

QListWidget on Windows, Mac & Ubuntu Linux.

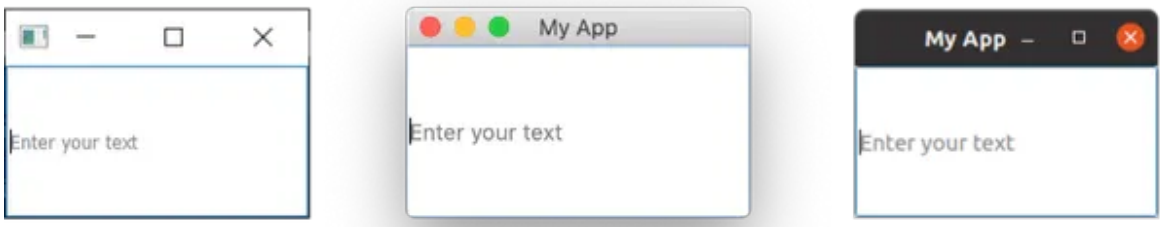
QLineEdit

The `QLineEdit` widget is a single-line text editing box, into which users can type input. These are used for form fields, or settings where there is no restricted list of valid inputs. For example, when entering an email address, or computer name:

PYTHON

```
class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()

        self.setWindowTitle("My App")

        widget = QLineEdit()
        widget.setMaxLength(10)
        widget.setPlaceholderText("Enter your text")

        #widget.setReadOnly(True) # uncomment this to make it read-only

        widget.returnPressed.connect(self.return_pressed)
        widget.selectionChanged.connect(self.selection_changed)
        widget.textChanged.connect(self.text_changed)
```

```
        widget.textEdited.connect(self.text_edited)

        self.setCentralWidget(widget)

    def return_pressed(self):
        print("Return pressed!")
        self.centralWidget().setText("BOOM!")

    def selection_changed(self):
        print("Selection changed")
        print(self.centralWidget().selectedText())

    def text_changed(self, s):
        print("Text changed...")
        print(s)

    def text_edited(self, s):
        print("Text edited...")
        print(s)
```



QLineEdit on Windows, Mac & Ubuntu Linux.

As demonstrated in the above code, you can set a maximum length for the text in a line edit using the `setMaxLength()` method.

The `QLineEdit` has a number of signals available for different editing events, including when the *Enter* key is pressed (by the user), and when the user selection is changed. There are also two edit signals, one for when the text in the box has been edited and one for when it has been changed. The distinction here is between user edits and programmatic changes. The `textEdited` signal is only sent when the user edits text.

Additionally, it is possible to perform input validation using an *input mask* to define which characters are supported and where. This can be applied to the field as follows:

```python
widget.setInputMask('000.000.000.000;_')
```

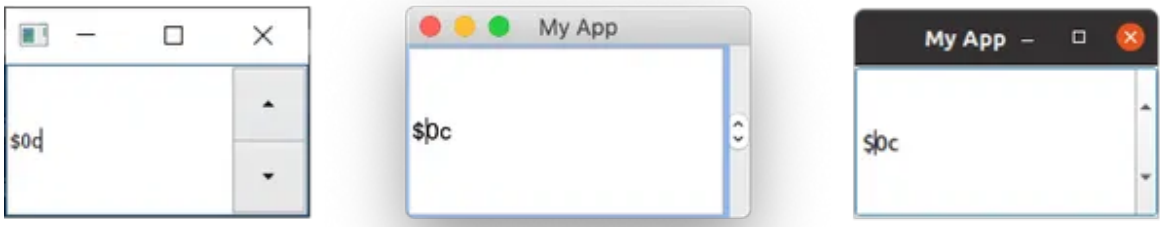
The above would allow a series of 3-digit numbers separated with periods, and could therefore be used to validate IPv4 addresses.

QSpinBox and QDoubleSpinBox

`QSpinBox` provides a small numerical input box with arrows to increase and decrease the value. `QSpinBox` supports integers, while the related widget, `QDoubleSpinBox`, supports floats:

```python
class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()

        self.setWindowTitle("My App")

        widget = QSpinBox()
        # Or: widget = QDoubleSpinBox()

        widget.setMinimum(-10)
        widget.setMaximum(3)
        # Or: widget.setRange(-10,3)

        widget.setPrefix("$")
        widget.setSuffix("c")
        widget.setSingleStep(3)  # Or e.g. 0.5 for QDoubleSpinBox
        widget.valueChanged.connect(self.value_changed)
        widget.textChanged.connect(self.value_changed_str)

        self.setCentralWidget(widget)

    def value_changed(self, i):
        print(i)

    def value_changed_str(self, s):
        print(s)
```

Run it, and you'll see a numeric entry box. The value shows pre and post-fix units and is limited to the range 3 to -10.



QSpinBox on Windows, Mac & Ubuntu Linux.

The demonstration code above shows the various features that are available for the widget.

To set the range of acceptable values, you can use the `setMinimum()` and `setMaximum()` methods. Alternatively, use `setRange()` to set both simultaneously. Annotation of value types is supported with both prefixes and suffixes that can be added to the number (e.g. for currency markers or units) using the `setPrefix()` and `setSuffix()` methods, respectively.

Clicking the up and down arrows on the widget will increase or decrease the value in the widget by an amount, which can be set using the `setSingleStep()` method. Note that this has no effect on the values that are acceptable to the widget.

Both `QSpinBox` and `QDoubleSpinBox` have a `valueChanged` signal, which fires whenever their value is altered. The raw `valueChanged` signal sends the numeric value (either an `int` or a `float`), while `textChanged` sends the value as a string, including both the prefix and suffix characters.

You can optionally disable text input on the spin box's line edit, by setting it to read-only. With this setting, the value can *only* be changed using the controls:

```python
widget.lineEdit().setReadOnly(True)
```

This setting also has the side effect of disabling the flashing cursor.

QSlider

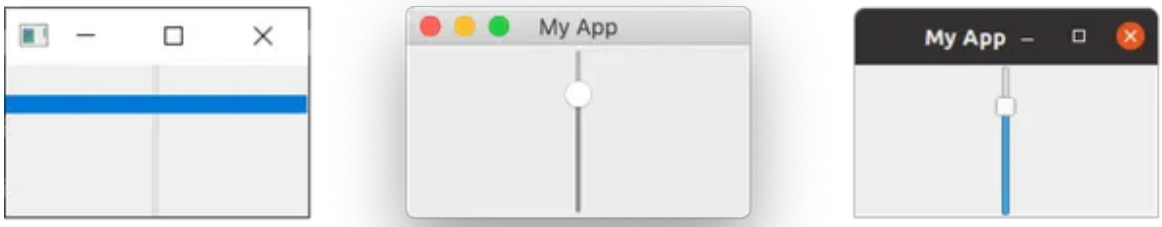
`QSlider` provides a slide-bar widget, which internally works like a `QDoubleSpinBox`. Rather than display the current value numerically, that value is represented by the position of the slider's handle along the length of the widget. This is often useful when providing adjustment between two extremes, but when absolute accuracy is not required. The most common use case of this type of widget is for volume controls in audio playback.

There is an additional `sliderMoved` signal that is triggered whenever the slider moves position and a `sliderPressed` signal that is emitted whenever the slider is clicked:

PYTHON

```
class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()

        self.setWindowTitle("My App")

        widget = QSlider()

        widget.setMinimum(-10)
        widget.setMaximum(3)
        # Or: widget.setRange(-10,3)

        widget.setSingleStep(3)

        widget.valueChanged.connect(self.value_changed)
        widget.sliderMoved.connect(self.slider_position)
        widget.sliderPressed.connect(self.slider_pressed)
        widget.sliderReleased.connect(self.slider_released)

        self.setCentralWidget(widget)

    def value_changed(self, i):
        print(i)

    def slider_position(self, p):
        print("position", p)

    def slider_pressed(self):
        print("Pressed!")

    def slider_released(self):
        print("Released")
```

Run this, and you'll see a slider widget. Drag the slider to change the value:



QSlider on Windows, Mac & Ubuntu Linux.

> The `singleStep` property of a `QSlider` only affects keyboard navigation when using arrow keys. When we interact with the slider using the mouse, the `singleStep` is ignored, and the slider can be moved freely.

You can also construct a slider with a vertical or horizontal orientation by providing the orientation as you create it. The orientation flags are defined in the `Qt` namespace:

```python
widget = QSlider(Qt.Vertical)
# Or:
widget = QSlider(Qt.Horizontal)
```

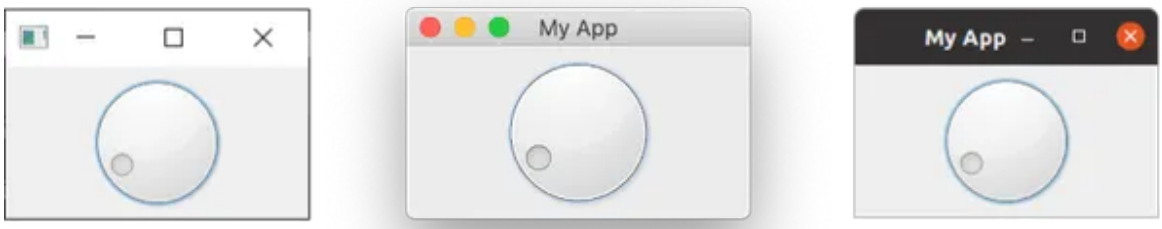
QDial

Finally, the `QDial` widget is a rotatable widget that works just like the slider but appears as an analog dial. This widget looks nice, but from a UI perspective, it is not particularly user-friendly. However, dials are often used in audio applications as a representation of real-world analog dials:

```python
class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()

        self.setWindowTitle("My App")

        widget = QDial()
        widget.setRange(-10, 100)
        widget.setSingleStep(0.5)
```

```
        widget.valueChanged.connect(self.value_changed)
        widget.sliderMoved.connect(self.slider_position)
        widget.sliderPressed.connect(self.slider_pressed)
        widget.sliderReleased.connect(self.slider_released)

        self.setCentralWidget(widget)

    def value_changed(self, i):
        print(i)

    def slider_position(self, p):
        print("position", p)

    def slider_pressed(self):
        print("Pressed!")

    def slider_released(self):
        print("Released")
```

Run this, and you'll see a circular dial. Rotate it to select a number from the range:



QDial on Windows, Mac & Ubuntu Linux.

The signals are the same as for the `QSlider` widget and retain the same names (e.g. `sliderMoved`).

Conclusion

This concludes our brief tour of the common widgets used in PyQt applications. To see the full list of available widgets, including all their signals and attributes, check out the Qt documentation.

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PyQt5 Widgets was written by Martin Fitzpatrick with contributions from Leo Well.

Martin Fitzpatrick has been developing Python/Qt apps for 8 years. Building desktop applications to make data-analysis tools more user-friendly, Python was the obvious choice. Starting with Tk, later moving to wxWidgets and finally adopting PyQt. Martin founded PythonGUIs to provide easy to follow GUI programming tutorials to the Python community. He has written a number of popular Python books on the subject.

PyQt5 Widgets was published in tutorials on May 05, 2019 (updated January 21, 2026) . Feedback & Corrections can be submitted here.

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